

Supervise, Intervene, Improve: Interface Design for Human Oversight of Multiple Autonomous Robot Manipulators

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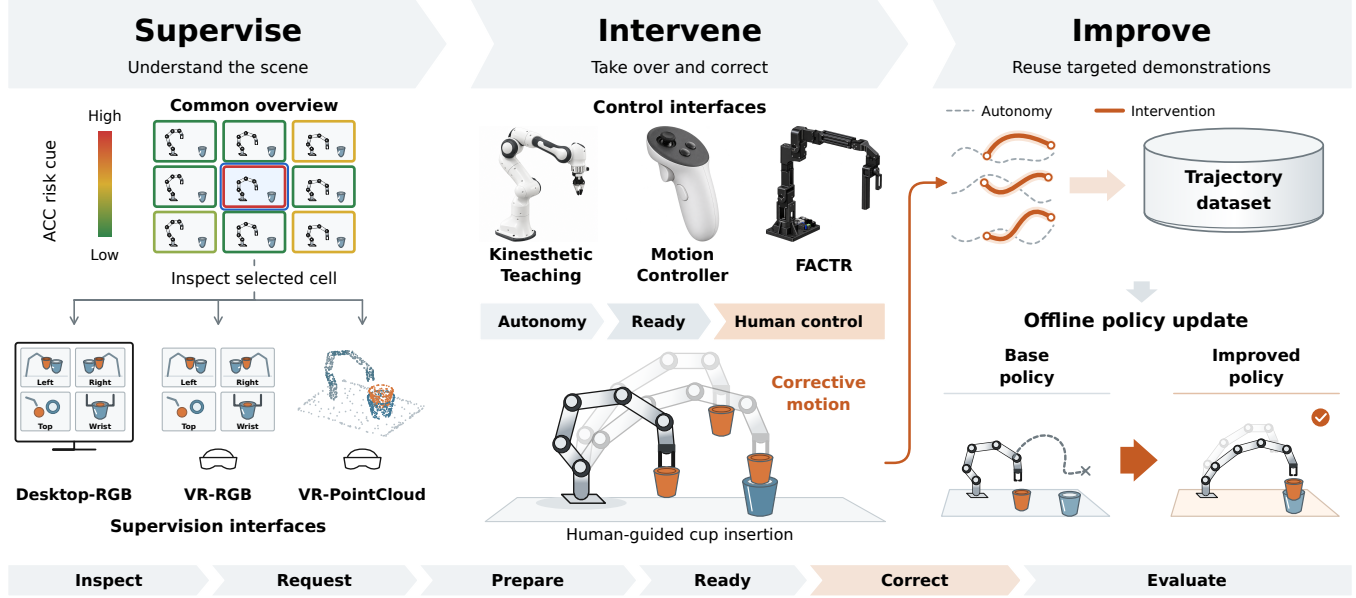


Figure 1: Selective-intervention workflow. Operators inspect a cell (blue selection outline) using one of three supervision interfaces; green-to-red borders show ACC risk (vertical scale). An intervention request pauses only that cell for controller preparation. Human control begins after Ready, while other unpaused cells continue autonomously. Ghosted robot poses illustrate human-guided cup insertion. Orange intervention segments are retained for offline policy updating. The base and improved policy scenes conceptually illustrate learning from corrections; all imagery and trajectories are illustrative.

Abstract

Selective human intervention allows one operator to support multiple autonomous manipulators while collecting corrective demonstrations for policy learning. Its effectiveness depends on how operators inspect robot scenes, provide corrections, and manage concurrent requests for assistance. We present Supervise–Intervene–Improve, a risk-informed framework linking supervision, corrective control, and intervention-data collection for offline learning. Three independent user studies with 72 participants evaluate visualization, corrective-control interfaces, and fleet size during simulated manipulation. We compare desktop RGB, VR RGB, and VR point-cloud visualization; kinesthetic teaching, motion controllers, and FACTR teleoperation; and fleets of one to nine robots. VR point-cloud visualization improved recovery and shortened corrections relative to both RGB interfaces. Kinesthetic teaching achieved higher recovery than motion controllers in the primary analysis, but required more total intervention time. Increasing fleet size reduced intervention coverage and delayed assistance, but increased successful-intervention throughput. These findings show that scalable human oversight requires interfaces that support effective corrections alongside timely access to assistance.

CCS Concepts

• Human-centered computing → Human computer interaction (HCI); • Computer systems organization → Robotics.

Keywords

multi-robot supervision, corrective intervention, virtual reality, teleoperation, human-in-the-loop learning

1 Introduction

Learned manipulation policies can perform complex behaviors, yet unfamiliar object configurations, imperfect grasps, and contact deviations can move them beyond the states represented in their training data. Selective human intervention allows autonomous execution to continue during routine operation, with an operator temporarily taking control when assistance is needed. These corrections can also provide targeted demonstrations for subsequent policy learning [10, 14, 17].

For one operator to support multiple manipulators, the interface must enable rapid scene understanding and effective corrective action while other robots continue operating [22, 23]. Local demonstration collection can rely on direct sight of the robot. Supervising distributed cells requires remote teleoperation, making

spatial understanding through mediated views a central challenge. Point-cloud telepresence offers a promising way to support this understanding [7]. Designing this workflow requires evidence about both the supervision interface used to inspect robot scenes and the corrective-control interface used to resolve problems. Scaling supervision further requires balancing correction throughput, intervention coverage, and response time. The learning value of the resulting demonstrations adds another design objective: effective assistance should also support improvements in future autonomy.

We present *Supervise–Intervene–Improve* (Fig. 1), a risk-informed framework connecting human supervision, selective correction, and offline policy learning. Simulated manipulators execute pre-trained Action Chunking Transformer (ACT) policies [34], held fixed during the user studies. Simulation provides controlled task conditions for comparing interfaces. A common Action Chunking Consistency (ACC) cue indicates disagreement among overlapping action predictions [28], while human operators decide which cell to inspect and when to intervene. Active corrections are recorded for subsequent policy updates.

Three independent within-subjects studies involve 72 participants, 24 per study. The Supervision Study compares Desktop-RGB, VR-RGB, and VR-PointCloud with kinesthetic teaching (KT) fixed. The Control Study compares KT, motion controllers (MC), and FACTR teleoperation with VR-PointCloud fixed. Prior work motivates this shared configuration [7, 16, 29]. The Fleet Study varies the number of active robots from one to nine with VR-PointCloud and KT fixed, examining how operator attention can support concurrent demand. An offline learning evaluation compares policies fine-tuned with corrections collected through the different control interfaces, assessing demonstration quality through subsequent autonomous performance.

We address four research questions: **RQ1**: How does the supervision interface affect recovery, spatial understanding, and workload? **RQ2**: How does the corrective-control interface affect recovery, intervention efficiency, and user experience? **RQ3**: How does fleet size affect correction throughput, intervention coverage, and response time? **RQ4**: How does the corrective-control interface affect the learning value of intervention data, measured by policy performance after fine-tuning?

Our contributions are a framework integrating selective intervention with synchronized demonstration capture; empirical comparisons of supervision and corrective-control interfaces; an evaluation of throughput and access to assistance across fleet sizes; and an assessment of how corrective demonstrations from different control interfaces support offline policy improvement. Together, these contributions provide empirical guidance for interfaces that support effective corrections, timely assistance, and subsequent policy learning for manipulators.

Project website: [ANONYMOUS PROJECT WEBSITE URL].

2 Related Work

2.1 Multi-Robot Supervision and Selective Intervention

Multi-robot supervision requires allocating attention among simultaneous autonomous processes. Fan-out and neglect tolerance characterize the relation between autonomous operation and human

servicing demands [24], while surveys describe how task structure and autonomy shape this interaction [6, 15]. Mixed-initiative control and coordinated shared autonomy address transitions in authority and the scheduling of human assistance [4, 8]. Our Fleet Study separates access to assistance from correction effectiveness.

2.2 Interfaces for Supervision and Corrective Control

Desktop, VR, and shared-control interfaces provide different ways to inspect and manipulate robot scenes [26, 31, 33]. Open-TeleVision, OPEN TEACH, and IRIS support immersive robot interaction and data collection [3, 11, 13]. A comparative telepresence study found point-cloud visualization to outperform the tested RGB-stream and Open-TeleVision configurations [7]. We compare matched RGB views across desktop/VR with an inspectable point cloud, varying configuration rather than isolating display modality.

Physical guidance enables kinesthetic teaching [1, 29], while GELLO [32] and FACTR [18] provide articulated leader devices for robot control and data collection. Comparative studies examine usability and demonstration quality [12, 16]. Li et al. [16] found advantages of kinesthetic demonstrations for tasks without strong contact forces, but users preferred teleoperation for large-scale collection because of physical effort and replay time. We examine these devices during brief corrections around autonomous execution, explicitly separating controller preparation from active correction.

2.3 Learning from Intervention

Dagger addresses the distribution shift between demonstrated and policy-visited states [27]. Human-gated and risk-aware approaches reduce expert involvement by selecting when assistance is provided [10, 14]. Human-in-the-loop imitation learning and deployment-time learning systems incorporate corrective experience into subsequent policy updates [17, 20, 30]. Physical corrections can also communicate desired behavior or objectives [19, 21]. We use an ACT policy [34] and a fixed ACC cue [28]; we do not propose a new failure detector. Our human studies connect access to assistance with correction quality. The offline protocol in Section 6 distinguishes targeted intervention data from simply adding successful autonomous experience.

3 Experimental Setup

One operator supervises multiple MuJoCo cells, each containing a simulated Franka Panda, task objects, virtual cameras, and a pretrained ACT policy [34]. Simulation provides repeatable scene configurations for interface comparisons and allows us to vary fleet size and introduce challenging situations systematically. Manipulation takes place in simulation, while corrections are provided through physical input devices. Policies remain fixed throughout the user studies.

3.1 Tasks and Success Detection

T-Shape Assembly and Cup Insertion (Fig. 2) combine grasping, relative-pose estimation, and precise placement with complementary geometric and visibility demands. T-Shape Assembly requires placing a T-shaped block on a base in an upright configuration, emphasizing orientation alignment and contact placement. Cup

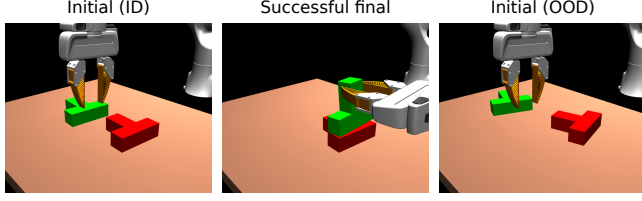
T-Shape Assembly**Cup Insertion**

Figure 2: Manipulation tasks and example evaluation states. Rows show T-Shape Assembly and Cup Insertion. Columns show an in-distribution initial state, a successful final state, and an out-of-distribution initial state.

Insertion requires inserting red into green and blue into yellow cups around occluding obstacles, emphasizing depth judgment and insertion alignment under partial occlusion. These properties allow us to examine how interfaces support scene understanding and corrective manipulation.

Success is determined from simulator state using task-specific geometric and contact criteria. T-Shape checks assess placement, alignment, contact, and gripper release. Cup checks require both designated pairs to satisfy insertion, alignment, and contact constraints, with cups and obstacles remaining upright. Detailed thresholds appear in Supplementary Section S12. This task-state detector determines completion, while the ACC cue described below supports decisions about assistance.

3.2 Nominal and Challenging Scenarios

We mix nominal randomized scenes with deliberately altered scenes to study selective assistance during routine and challenging autonomous execution. We use in-distribution (ID) and out-of-distribution (OOD) as operational labels for these scenario banks. They distinguish scene-generation conditions rather than episode outcomes. Either bank can contain episodes that succeed autonomously or need correction.

In Supervision and Control, ID scenes perturb object positions independently by up to 1 cm along each horizontal axis. T-shaped blocks and boxes also receive yaw perturbations of up to 5°, while cup yaw is unchanged. OOD T-Shape scenes jointly alter both blocks' positions and headings. OOD Cup scenes change cup heights and exchange the two boxes' textures while retaining nominal placement randomization. These changes introduce geometric and visual variation for evaluating corrective assistance. Object-specific ranges appear in Supplementary Section S12.

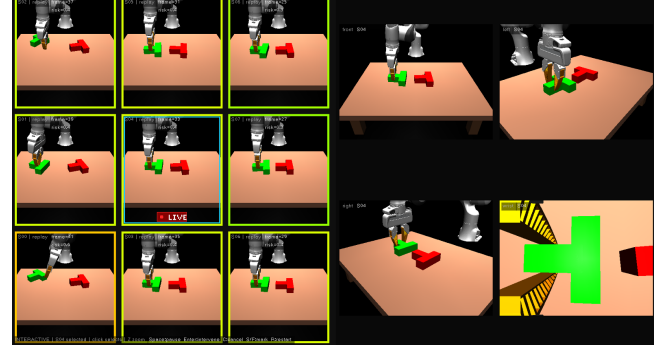


Figure 3: Desktop-RGB operator view during T-Shape Assembly. The left panel shows nine concurrent cells with ACC risk cues along their borders. The right panel shows four enlarged views of the selected cell. Inspection leaves autonomy running until the operator requests intervention.

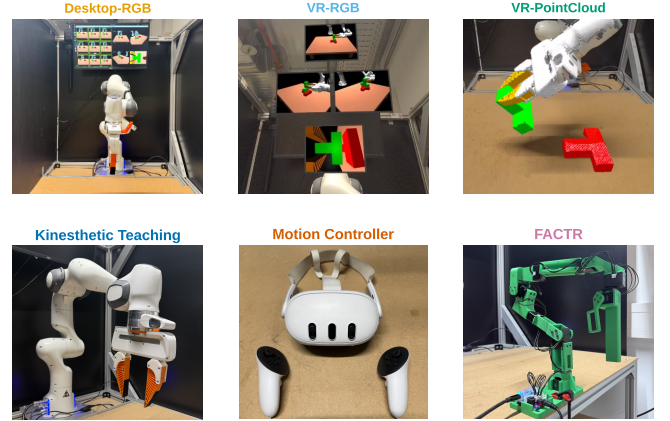


Figure 4: Supervision and corrective-control interfaces. Top row, left to right: Desktop-RGB, VR-RGB, and VR-PointCloud, illustrated during T-Shape Assembly. Bottom row: the physical Panda used for Kinesthetic Teaching, the Meta Quest headset and controllers used for Motion Controller interaction, and the FACTR articulated leader.

3.3 Interfaces and Intervention Workflow

A shared overview displays all cells with a green-to-red Action Chunking Consistency (ACC) cue indicating disagreement among overlapping policy predictions [28] (Fig. 3). Operators decide which cell to inspect and when to intervene.

These interfaces model remote supervision by mediating the operator's view of the task scene. Desktop-RGB presents enlarged left, right, top, and wrist camera views after cell selection. VR-RGB presents the same views in a Meta Quest 3. VR-PointCloud presents fused geometry registered to the simulated workspace (Fig. 4), supporting spatial inspection through 3D telepresence [7]. The overview and workflow are shared across conditions, with other cells remaining visible. VR passthrough lets operators see the physical control devices.

Kinesthetic Teaching (KT) uses a physical Panda aligned to the selected simulated robot before gravity-compensated guidance commands its motion. FACTR similarly aligns an articulated leader for physical guidance [18]. Motion Controller (MC) maps tracked translation and rotation to relative end-effector motion, with buttons controlling the gripper. Physical alignment contributes to controller preparation time.

Inspection leaves autonomy running. Requesting intervention pauses the selected cell for controller preparation and subsequent human correction. Only one cell receives human control at a time, while other unpaused cells continue autonomously. The task-state detector is inactive during human control. After release, it checks completion before another autonomous action. A terminal outcome ends the episode, and autonomy may otherwise resume. Active corrections are retained for offline policy learning.

4 User Studies

4.1 Design and Participants

We conducted three within-subjects studies with independent cohorts of 24 participants each (72 total). Each participant experienced all conditions of one study (Table 1). Participants were recruited through university advertisements and campus flyers, provided written informed consent, and were offered EUR 15 or study credit for approximately 90-min sessions. Supplementary Table S8 reports age.

4.2 Counterbalancing and Procedure

Participants received a briefing, completed comprehension checks on the ACC cue, robot states, takeover, release, and safety, and practiced until completing at least one successful intervention. Each study prescribed 30 min of measured interaction. Supervision and Control allocated 10 min per interface, split equally between Cup Insertion and T-Shape Assembly. Fleet allocated 7.5 min to each of four fleet sizes. Questionnaires followed each condition, with comparative ratings at the end. Briefing, practice, questionnaires, and breaks were excluded from measured interaction time.

Each of the six interface orders occurred four times per Supervision/Control cohort. Twelve participants performed each task first. Task order was associated with interface-order group. Fleet used all 24 condition-order permutations once. F1, F3, F6, and F9 denote the number of active cells. Further procedural details appear in Supplementary Sections S9 and S11.

Supervision and Control operated nine cells concurrently and replenished OOD scenes to maintain two to three challenging cases. Fleet targeted 0.4 scheduled incidents per robot-minute. Scheduling, scene activation, and the onset of assistance need are distinct events (Supplementary Section S12).

4.3 Outcomes and Eligibility

In Supervision and Control, *Recovery* measures whether a correction enables task completion, either at the first valid post-release check or during subsequent autonomous execution in the same cell and episode. A terminal failure or another intervention request before success ends the attempt unsuccessfully. Outcomes not established before observation ends remain unresolved. This

shared endpoint was harmonized retrospectively (Supplementary Section S12).

Preparation delay runs from intervention request to READY, which marks preparation completion. The logged human-control-start event begins active correction, which ends at release. Total intervention time runs from request to release and includes any Ready-to-start interval. These timings exclude pre-request inspection and response time.

Fleet measures how broadly and promptly assistance is provided. *Coverage* is the proportion of recorded assistance needs that reach human-control start. *Response delay* measures the wait from need onset to human-control start among assisted needs. An effective need is identified by an uncanceled online T_{need} annotation for a cell and episode. Needs still unserved at the observation boundary remain uncovered. Timing uses the recorded need onset, rather than the incident's scheduled time.

Useful intervention throughput counts tasks completed at the first valid post-release check per minute of condition time. It therefore measures immediate completion, whereas Supervision/Control Recovery also includes later autonomous completion.

Agreement items use seven-point scales. Interface rankings and Fleet forced-choice preferences are analyzed separately from objective performance. All six NASA-TLX dimensions [9] use a 0–20 scale. Higher Performance ratings indicate better performance, from Failure (0) to Perfect (20). Higher ratings on the other five dimensions indicate greater workload. No weighted composite is used.

Attempts lasting 1.5 s or less are excluded to reduce the influence of brief accidental activations. After all attempt exclusions, Supervision retained 708 of 740 requests and Control retained 910 of 972. Recovery models use 631 and 857 known outcomes, respectively, with 77 and 53 unresolved outcomes left missing. All 24 participants in each cohort contribute data. Fleet analyses use 95 eligible full blocks. Detailed exclusions, outcome-specific denominators, and sensitivity analyses appear in Supplementary Sections S4 and S12.

4.4 Statistical Analysis

Studies are analyzed separately using Python and R. Supervision/Control objective models adjust for condition, task, and condition order. Task order was associated with interface-order group and was not included as a separate predictor. Recovery uses logistic mixed models with condition-by-task interactions and participant/scenario random intercepts. Task-adjusted estimates give both tasks equal weight, preventing tasks with more attempts from dominating the comparison. Positive times use log-time Gaussian models, reported as adjusted geometric means and time ratios. Counts use exposure-offset Poisson models. Fleet models account for condition position and participant dependence. Censored response timing additionally uses participant-clustered Cox inference.

Ordinal ratings and rankings use Friedman tests, Kendall's W , and paired Wilcoxon follow-ups. NASA-TLX uses participant-intercept Gaussian models. We report unadjusted 95% confidence intervals and Holm-adjusted p_H . Objective pairwise comparisons are corrected within outcome, TLX comparisons across dimensions, and ordinal/ranking families separately. Task interactions, personalization, and concurrency associations are exploratory. Omnibus

Table 1: Independent within-subjects study designs ($n = 24$ each). Supervision and Control include both tasks. Their primary objective estimates give both tasks equal weight.

Study	Varied factor	Fixed configuration
Supervision	Desktop-RGB, VR-RGB, VR-PointCloud	KT, nine cells, both tasks
Control	KT, MC, FACTR	VR-PointCloud, nine cells, both tasks
Fleet	F1, F3, F6, F9	KT, VR-PointCloud, T-Shape Assembly

rating tests establish overall differences without identifying specific pairs. Non-significance does not establish equivalence. Model specifications, simplifications, diagnostics, software details, and full inference appear in Supplementary Sections S4, S10, and S11, and Tables S1 and S3–S7.

5 User-Study Results

Table 2 summarizes post-hoc selected contrasts; Figures 5–9 show the separate outcomes.

5.1 Supervision Study

Recovery and correction time. Recovery was highest with VR-PointCloud (83.6%), followed by Desktop-RGB (68.4%) and VR-RGB (56.0%). All three pairwise contrasts were significant after Holm correction (Fig. 5a,d; Table 2). VR-PointCloud also enabled shorter active corrections (28.9 s) than Desktop-RGB (43.3 s) and VR-RGB (46.4 s). Both timing differences survived correction; the difference between the RGB interfaces was inconclusive (Fig. 5b; Supplementary Table S5).

Spatial understanding. Ratings differed across representations (Friedman $\chi^2(2) = 19.75$, $W = 0.411$, $p_H < .001$), with descriptively highest ratings for VR-PointCloud. This omnibus test does not by itself establish both pairwise differences. Relative-pose judgment also differed ($\chi^2(2) = 34.80$, $W = 0.725$, $p_H < .001$). Confidence in correction sufficiency ($W = 0.231$, $p_H = .015$) and visual comfort ($W = 0.310$, $p_H = .003$) also differed in corrected Friedman tests. Attention identification, situation awareness, and risk-cue usefulness did not differ detectably (Supplementary Table S3). These ratings do not measure missed or unnecessary interventions.

5.2 Control Study

Recovery and efficiency. Recovery was 88.9% for KT, 78.3% for MC, and 84.7% for FACTR (Fig. 6). KT exceeded MC; neither FACTR recovery contrast survived correction. KT and MC corrected faster than FACTR, without a detectable KT/MC difference (Table 2; Supplementary Table S6).

MC's near-zero request-to-Ready interval reflects software setup; it excludes pre-request response and any Ready-to-control-start delay. KT prepared faster than FACTR (ratio = 0.797, 95% CI [0.736, 0.863], $p_H < .001$), while KT's total intervention time exceeded MC's (ratio = 1.109, 95% CI [1.019, 1.208], $p_H = .017$).

Perceived control and precision. The corrected Friedman tests detected differences in mapping ($\chi^2(2) = 10.53$, $W = 0.219$, $p_H = .031$) and precision ($\chi^2(2) = 13.58$, $W = 0.283$, $p_H = .008$), but not perceived control ($\chi^2(2) = 7.75$, $W = 0.162$, $p_H = .104$). Item follow-ups appear in Supplementary Table S6.

Sensitivity checks. The primary analysis found higher Recovery with KT than MC. Evidence for this contrast was weaker in sensitivity analyses and did not reach the corrected significance threshold when outcomes were summarized per participant ($p_H = .055$) or when short attempts were retained ($p_H = .066$). Both checks remained directionally consistent with the primary result, but the statistical evidence for the KT–MC difference was sensitive to these analysis choices. These analyses should not be interpreted as evidence of equivalence between the controllers. Timing diagnostics and the separate three-controller omnibus analysis are reported in Supplementary Sections S4 and S12.

5.3 Fleet Study

Throughput and coverage. Throughput rose descriptively, with similar F6/F9 estimates (Fig. 7), but the omnibus test was inconclusive ($\chi^2(3) = 7.22$, $p = .065$). F9/F1 also fails Holm correction (Table 2); the apparent plateau is descriptive.

Coverage decreased across fleet sizes ($\chi^2(3) = 309.16$, $p < .001$). All pairwise contrasts survived correction, including F3/F1 (OR = 0.051, 95% CI [0.023, 0.111], $p_H < .001$) and F9/F6 (OR = 0.605, 95% CI [0.486, 0.753], $p_H < .001$). This deterioration begins between F1 and F3.

Delay and correction outcomes. Among intercepted needs, F3/F1 need-to-control delay had ratio = 2.678, 95% CI [2.218, 3.235], $p_H < .001$; F6 and F9 also exceeded F1, while comparisons among larger configurations were inconclusive after correction. The censored timing analysis also detected a fleet effect, with a borderline fleet-term proportional-hazards diagnostic (Supplementary Section S4).

Active-correction duration did not differ detectably ($\chi^2(3) = 5.02$, $p = .171$). This does not establish equal correction times or describe needs that never reached control. Additional post-release outcome diagnostics appear in the Supplementary Material.

5.4 Workload and Preferences

Figures 8 and 9 separate workload from rankings and preferences ($n = 24$ per question/cohort). In Supervision, VR-PointCloud had lower Mental Demand, Physical Demand, Temporal Demand, Effort, and Frustration than both RGB conditions in corrected pairwise tests; no Performance pair survived correction (Supplementary Table S5). For example, Mental Demand was lower under VR-PointCloud than Desktop-RGB (Desktop minus PointCloud: difference = 6.333, 95% CI [4.710, 7.957], $p_H < .001$) and VR-RGB (difference = 7.375, 95% CI [5.752, 8.998], $p_H < .001$), in scale points. VR-PointCloud received 95.8% of overall first ranks.

In Control Study, KT reduced Mental Demand relative to MC and FACTR, and Frustration relative to MC (Supplementary Table S6). No other TLX pair survived correction, including Performance.

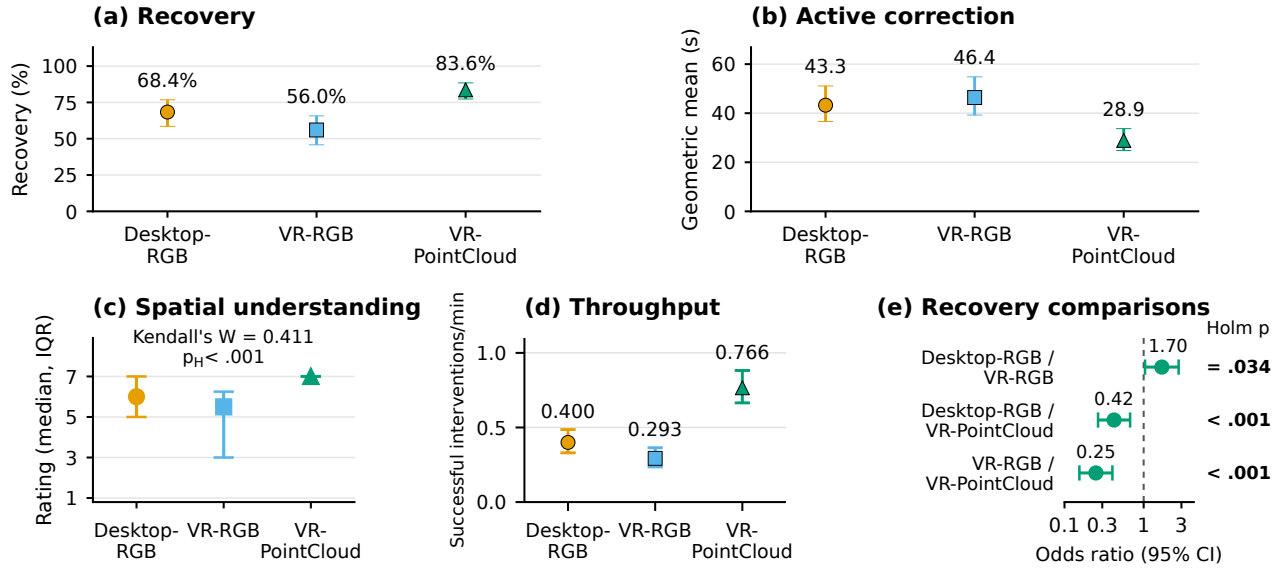


Figure 5: Supervision ($n = 24$; 708 retained attempts, 631 known Recovery outcomes). (a,b) Recovery and active-correction time with 95% CIs, estimated with equal task weights. Recovery is task success after release or during autonomous continuation in the same episode, before a new intervention request. (c) Spatial-understanding median/IQR (1–7), Kendall's W , and Holm-adjusted Friedman p_H . (d) Recovery ORs and 95% CIs; filled points indicate $p_H < .05$. Higher Recovery and ratings indicate better outcomes; shorter times indicate faster interventions. Accounting and sensitivity analyses: Supplementary Section S12.

Table 2: Selected model contrasts in independent cohorts ($n = 24$ each). Contrasts divide the first condition by the second. Recovery is task success after release or subsequent autonomous continuation before another request. OR: odds ratio; Ratio: time ratio. Intervals are unadjusted 95% CIs; significance requires Holm $p_H < .05$. Full recovery contrasts: Supplementary Section S12; other contrasts: Tables S5–S7.

Study	Outcome	Contrast	Effect	95% CI	Holm p
Supervision	Recovery	Desktop-RGB / VR-PointCloud	OR 0.423	[0.265, 0.674]	$p_H < .001$
Supervision	Recovery	VR-RGB / VR-PointCloud	OR 0.249	[0.154, 0.403]	$p_H < .001$
Control	Recovery	KT / MC	OR 2.212	[1.393, 3.511]	$p_H = .002$
Control	Active time	KT / FACTR	Ratio 0.831	[0.752, 0.918]	$p_H < .001$
Control	Active time	MC / FACTR	Ratio 0.838	[0.760, 0.925]	$p_H < .001$
Fleet	Coverage	F3 / F1	OR 0.051	[0.023, 0.111]	$p_H < .001$
Fleet	Throughput	F9 / F1	Rate ratio 1.316	[1.059, 1.636]	$p_H = .080$

Overall first ranks were KT 41.7%, MC 54.2%, and FACTR 4.2%. KT led first ranks for precision, mapping, naturalness, and repeated use (Fig. 9). Complete overall rankings were inconclusive ($W = 0.109$, $p_H = .072$); first-rank shares are descriptive. Fleet F9 exceeded F1 in Effort (difference = 3.875, 95% CI [2.172, 5.578], $p_H < .001$) and Temporal Demand (difference = 2.792, 95% CI [1.153, 4.430], $p_H = .028$). F3 was most often preferred for sustainable supervision (41.7%) and overall balance (54.2%).

6 Learning from Selective Interventions

We evaluate whether incorporating collected corrections improves subsequent autonomous performance (Fig. 10). The planned offline comparisons use held-out ID and OOD scenarios for both tasks; only active correction segments are retained.

[Learning results pending: insert verified data budgets, source-disjoint splits, seeds, success estimates and 95% CIs here; report matched-budget and all-available-data comparisons separately.] Supplementary Fig. S6 reserves detailed result slots. No policy improvement is claimed before completed runs are verified.

7 Discussion

7.1 Representation and Corrective Control

VR-PointCloud improved task-averaged recovery, correction time, and five NASA-TLX dimensions relative to both RGB representations. Exploratory recovery contrasts favored it for Cup Insertion; all T-Shape contrasts were inconclusive (Supplementary Section S14). These findings extend point-cloud telepresence research [7] to selective correction. Geometry may aid pose judgments, but display modality and representation were not isolated.

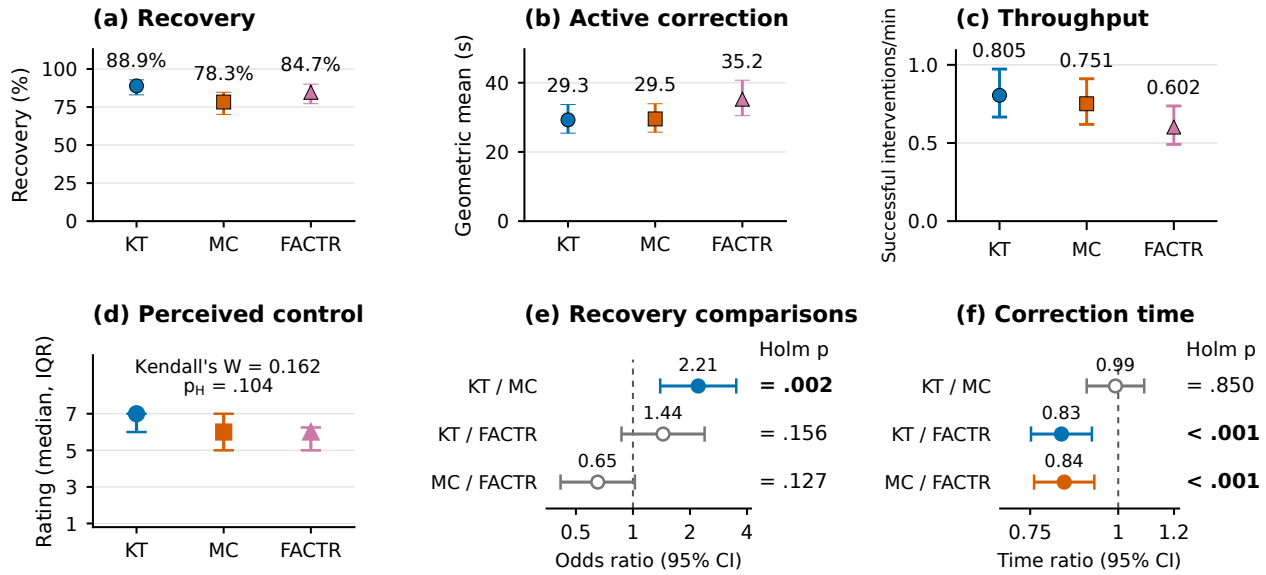


Figure 6: Control ($n = 24$; 910 attempts, 857 known recovery outcomes). (a–c) Equal-weight task-adjusted logistic/log-time estimates and 95% CIs. (d) Control-rating median/IQR, Kendall's W , and Holm-adjusted Friedman p_H . (e,f) Ratios and 95% CIs; filled points: $p_H < .05$. Higher Recovery and ratings indicate better outcomes; shorter times indicate faster interventions. MC preparation records software setup latency; it excludes the operator's response time.

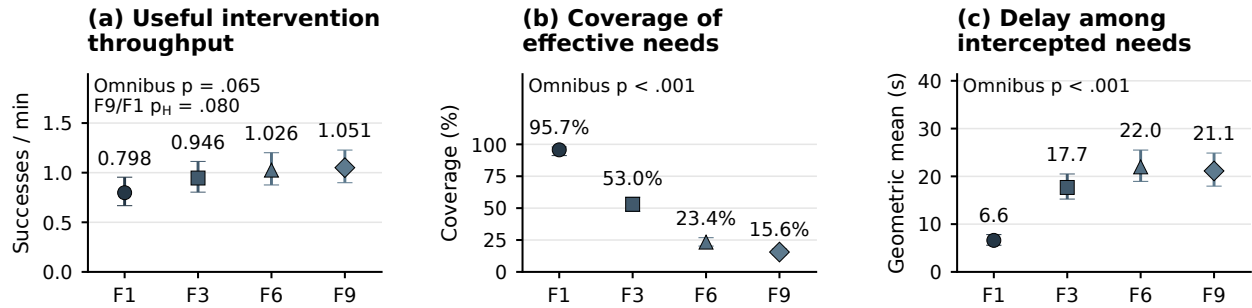


Figure 7: Fleet Study ($n = 24$, 95 primary blocks): model estimates and 95% CIs with omnibus tests. (a) Useful successes/minute: Poisson model. (b) Coverage of 2663 needs: logistic model. (c) Delay among 756 interceptions: log-time model. Models retain position/participant dependence. Higher throughput/coverage and shorter delay are favorable. F9/F1 throughput remains inconclusive after Holm correction.

Selected-attempt outcomes and inconclusive awareness ratings do not establish better failure detection.

KT favored recovery over MC in the primary model, consistent with prior work on physical guidance of virtual robots [12]. However, the participant-summary and short-attempt comparisons were inconclusive. Physical alignment also adds preparation cost: KT required more total intervention time than MC. Controller evaluation must therefore consider post-release outcomes alongside preparation and correction time.

7.2 Performance and Subjective Choice

In Supervision Study, Recovery, correction time, and descriptive preference favored VR-PointCloud. In Control Study, KT reduced

Mental Demand and Frustration relative to MC and led criterion-specific first ranks; MC received the most overall first ranks. Complete rankings and perceived-control ratings were inconclusive. Prior work likewise distinguishes kinesthetic intuitiveness from teleoperation preference [16]. Preparation overhead could contribute to preference here, but its causal role was not tested.

7.3 Access to Assistance

Larger fleets reduced coverage and delayed assistance, consistent with supervisory-control and fan-out accounts [2, 5, 24, 25]. Completing useful corrections can therefore coexist with substantial unmet demand. Throughput alone does not indicate whether assistance reaches all robots needing it.

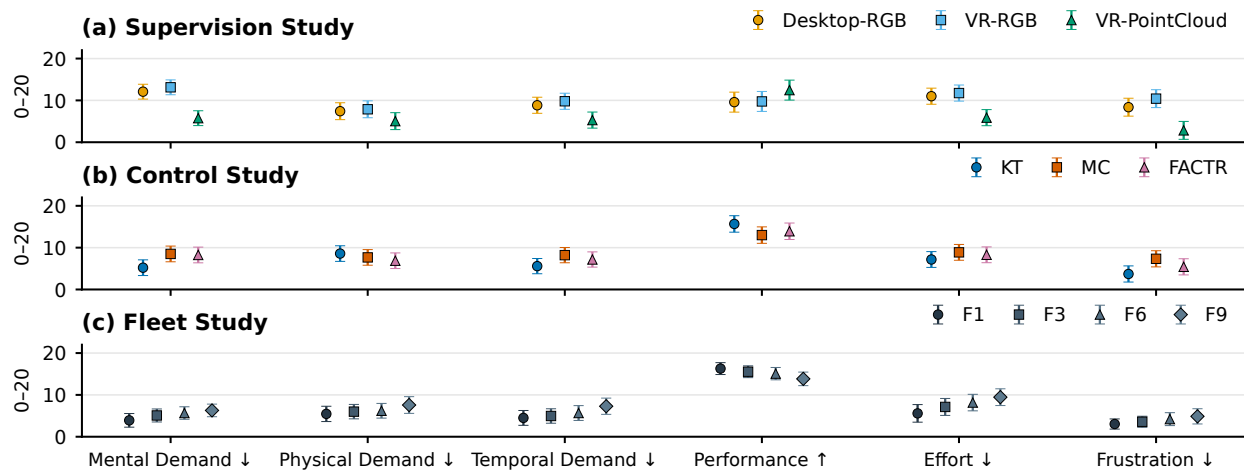


Figure 8: NASA-TLX ($n = 24$ per independent cohort): Gaussian mixed-model estimates and 95% CIs, 0–20. Favorable direction: ↑ higher, ↓ lower. Performance: 0=Failure, 20=Perfect; other dimensions increase with workload. Within-study Holm correction covers pairwise condition comparisons across all six dimensions (Supplementary Tables S5–S7).

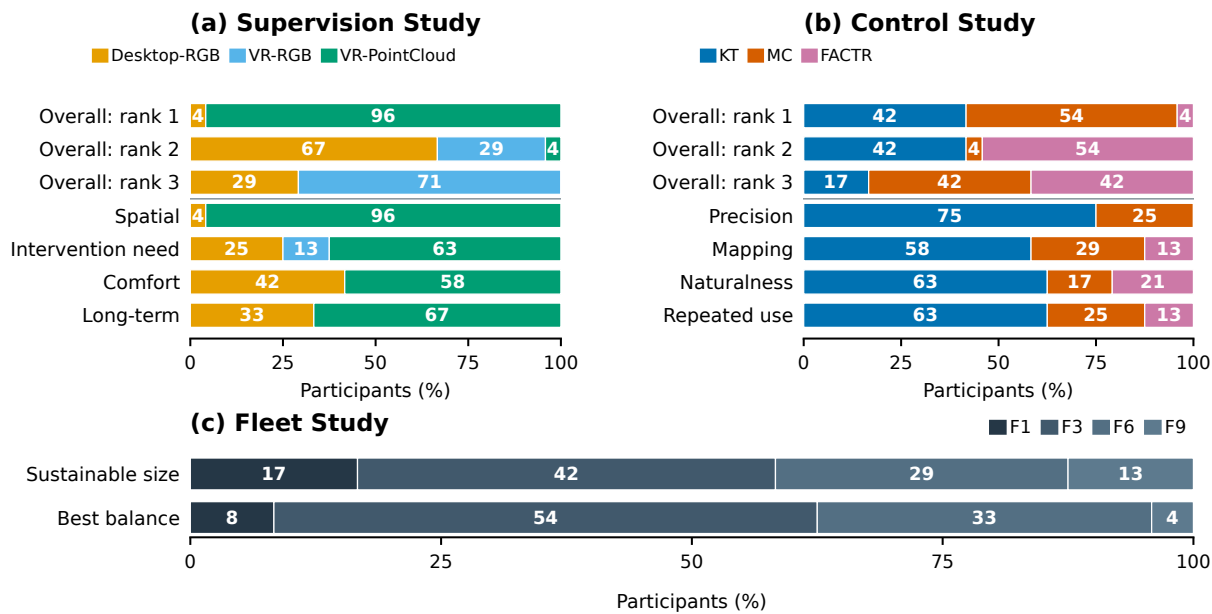


Figure 9: Interface rankings and supervision preferences. Each question includes 24 respondents from its respective independent cohort. (a,b) Overall ranking distributions and the interface ranked first for each criterion. Rank 1 denotes the most preferred interface. (c) Fleet-size choices for sustainable supervision and best overall balance.

The coverage decline between F1 and F3 does not define a universal capacity threshold: robot count changes both visual burden and total demand. Preference for a larger fleet does not establish overconfidence because the questions did not measure predicted coverage.

7.4 From Corrections to Learning Data

Deployment-time interventions can support policy updates [17], but successful recovery does not establish learning value. Matched-budget rollouts must distinguish correction value from additional experience (Fig. 10); results remain pending.

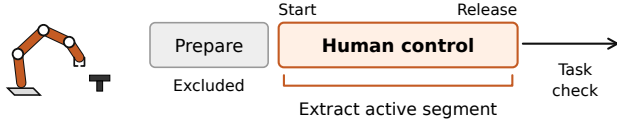
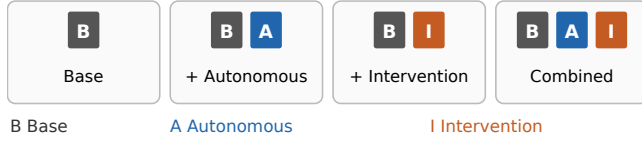
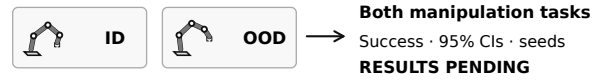
(a) Retain the human correction**(b) Compare training-data sources****(c) Evaluate held-out autonomy**

Figure 10: Offline learning protocol. (a) Extract active corrections, excluding preparation. (b) Compare planned training-data combinations. (c) Evaluate both tasks on held-out ID/OOD scenarios. Results, episode/seed counts and 95% CIs remain pending; no measured learning result is shown.

8 Limitations

Manipulation was simulated with physical KT/FACTR devices. The interface studies cover two tasks; Fleet covers one task and demand policy. Longer use may change fatigue and preference. Independent cohorts and a nonfactorial design preclude supervision-by-controller interaction claims and cross-study recovery comparisons. Unresolved outcomes vary by condition and task and may be informatively censored. Fleet release defaults and demand-label mixtures varied. Throughput tests remained inconclusive after excluding earlier releases or their participant; exact runtime overrides remain undocumented (Supplementary Section S13). Log-time diagnostics, aggregation, and short-attempt sensitivity qualify inference. Offline learning claims require verified datasets and evaluation distributions.

9 Conclusion

Supervise–Intervene–Improve connects risk-informed supervision, selective correction, and intervention-data capture for offline learning. Across three independent studies, VR-PointCloud supported higher task-averaged Recovery and faster correction than RGB representations. Kinesthetic Teaching favored Recovery over Motion Controller in the primary analysis, while preparation costs and sensitivity checks qualified that advantage. Larger fleets reduced coverage and delayed assistance, showing that successful corrections alone do not ensure timely support for every robot. Effective oversight therefore requires interfaces that support scene understanding and corrective control while accounting for operator time and access to assistance. Whether the collected corrections improve subsequent autonomous performance remains an open empirical question.

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